Appendix 1. Imprecision rating when the target of certainty rating is a small, moderate or large effect

The principles in this appendix for imprecision judgment when the target of certainty rating is a small, moderate or large effect are modified from the GRADE guideline 35.[1]

When Core GRADE users are interested in whether an effect is small, moderate, or large, they would rate their certainty in relation to a range of small effect (a range between the MID and the moderate effect threshold), moderate effect (a range between the moderate and large effect threshold), or large effect (greater than the large effect threshold) (Figure 1 in the main text).

In such situations, for imprecision ratings, the primary criterion is still judging whether the confidence interval crosses the threshold at either end of the range of interest. If yes, Core GRADE users rate down for imprecision. If the confidence interval appreciably crosses the range (e.g., crosses both thresholds that form the range), they consider rating down two levels for imprecision.

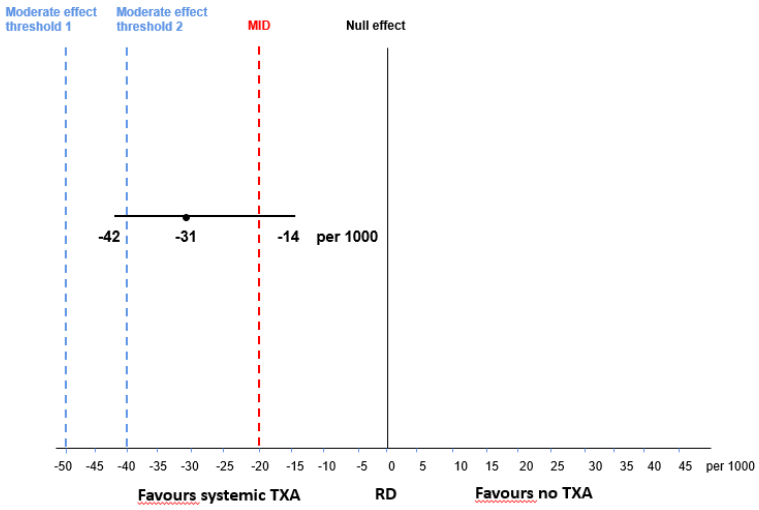
Consider a systematic review of tranexamic acid (TXA) versus no TXA (placebo or usual care) for patients receiving a percutaneous nephrolithotomy.[2] A meta-analysis of randomized controlled trials revealed that TXA yielded a reduction in blood transfusion by 31 fewer per 1000 with a confidence interval from 42 fewer to 14 fewer (Figure 1).

If Core GRADE users were interested in the magnitude of effect, they would have set thresholds, say an MID threshold at 20 fewer per 1000 and the moderate effect threshold at 50 fewer per 1000 (Moderate effect threshold 1 in Appendix 1 Figure 1) (i.e., a range of small effect between 20 to 50 per 1000). As the point estimate falls within the range of small effect, they would rate certainty that compared with no TXA, TXA has a small but important reduction in blood transfusion. Because the confidence interval also includes a trivial effect (i.e., the confidence interval crosses the MID threshold), they would rate down one level for imprecision.

If they consider that blood transfusion is more important to patients and set the moderate effect threshold at 40 per 1000 (Moderate effect 2 in Appendix 1 Figure 1), they can still rate certainty that a small effect is present. As the confidence interval now includes a moderate effect and a trivial effect, they might consider rating down two levels for imprecision.

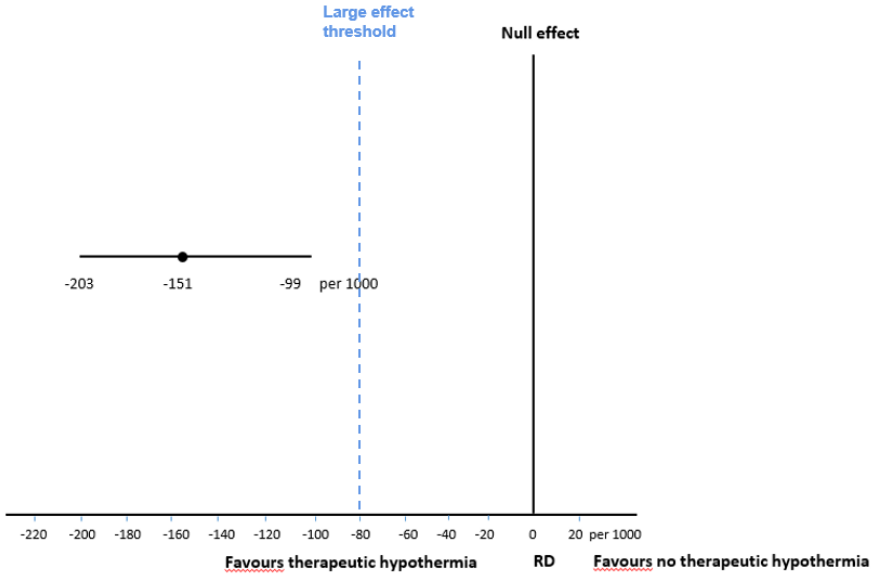
GRADE plain language can also help judge imprecision ratings when the target of certainty rating is a small, moderate, or large effect. Assume there are no concerns on the other four GRADE domains. In this case, when the confidence interval includes both a moderate effect and a trivial effect, Core GRADE users need to consider whether it is more appropriate to convey the message that a) TXA likely has a small but important reduction on blood transfusion" (i.e., a statement that would accompany rating down once) or b) that "TXA may have a small but important effect" (i.e., a statement that would accompany rating down twice). If the authors found the latter statement more appropriate, they would rate down twice for imprecision.

Figure 1. Effect of tranexamic acid versus no tranexamic acid on blood transfusion in patients who received percutaneous nephrolithotomy



Consider another systematic review of cooling for newborns with hypoxic ischaemic encephalopathy.[3] A meta-analysis of randomized controlled trials showed that therapeutic hypothermia yielded 151 fewer deaths or major neurodevelopmental disability per 1000 patients with a confidence interval from 203 fewer to 99 fewer per 1000. Had the review authors been interested in whether a large effect exists, they would rate certainty in relation to a large effect threshold. Considering the importance of the combined outcome (death or major neurodevelopmental disability), had the authors set the large effect threshold at a difference of 80 per 1000, they would rate their certainty that therapeutic hypothermia has a large effect. As the entire CI represents a large effect, they would not rate down for imprecision.

Figure 2. Effect of therapeutic hypothermia versus no therapeutic hypothermia on death or major neurodevelopmental disability in newborns with hypoxic ischaemic encephalopathy.



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Appendix 2. Choosing between random effects and fixed effect models

An important choice Core GRADE users conducting systematic reviews will face is whether, in their analysis, to use fixed effect or random effects statistical models. Fixed effect ignores variability in results across studies. Random effects considers that variability and generates wider confidence intervals the greater the variability.

Thus, the key points to know about the choice of model is the following:

i) When little heterogeneity exists (I^2 at or near 0) fixed effect and random effects models will yield very similar results.

ii) With substantial heterogeneity there will be two differences between the models:
   A) The confidence interval will be wider in random models
   B) If the small and larger studies yield different results the point estimate in the fixed will reflect the results in the larger studies to a greater extent than random models.

Statisticians disagree about the merits of the two approaches and there is no right answer regarding which to use. An article suitable for Core GRADE users provides details about the approach [1] and we present the key issues below.

When heterogeneity is substantial, the models will yield different results. In particular, confidence intervals will typically be wider in random effects and the gradient in weight between large and small studies will be greater in fixed effect models. The result will be that point estimates in fixed effect models will be closer to the large study results than point estimates in random effects model.

The figure below presents three hypothetical meta-analyses.[1] In A there is very little variability and so fixed and random models yield the same pooled estimates. In B there is considerable variability and so the random model produces a considerably wider confidence interval. However, because all studies are a similar size and weight (depicted by the same width confidence interval) the point estimates from the two models are identical. In C there is again high heterogeneity and so the confidence interval is wider for the random model. Here though, the small and large studies have different results and so the random point estimate is closer to the small studies than the fixed.

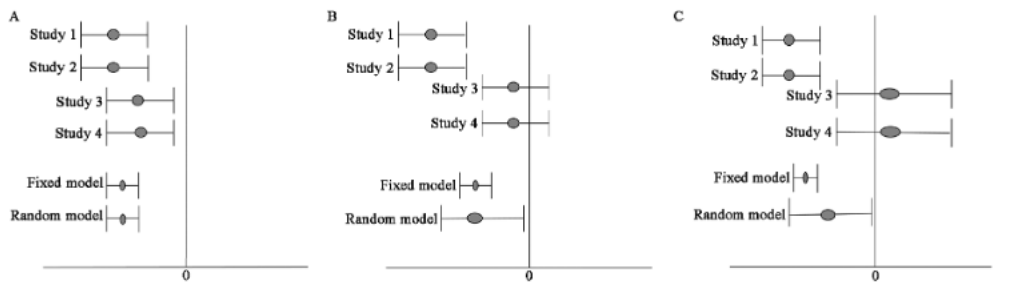

Figure 1: Hypothetical examples. (A) Minimal variability: Fixed and random models yield identical results. (B) The effect of model choice on precision. (C) The effect of model choice on the point estimate and precision.

The figure below presents actual meta-analysis illustrating the three situations: A with very little variability and so the same pooled estimates; B with large variability and so wider confidence intervals with random models but because of similar magnitude of effect in large and small studies very similar point estimates; and C with both large variability and difference in magnitude of effects between large and small studies and so differences in both pooled point estimates and width of their confidence intervals.

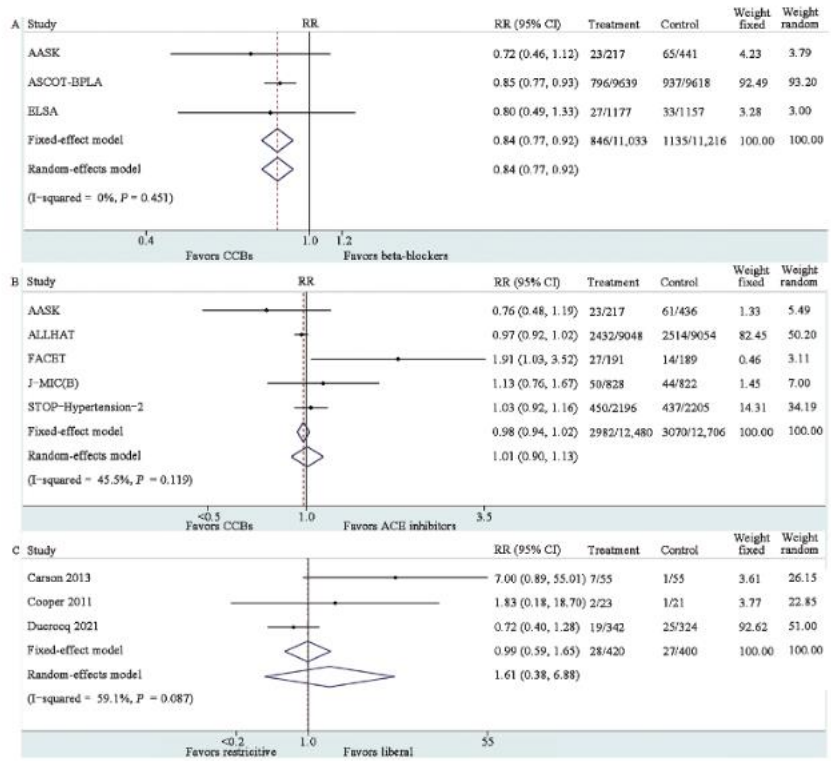

Figure 2: Real-world examples. (A) Meta-analysis comparing CCBs and beta-blockers on major cardiovascular events in patients with hypertension. (B) Meta-analysis comparing CCBs and ACE inhibitors on major cardiovascular events in patients with hypertension. (C) Meta-analysis comparing restrictive and liberal transfusion on 30-day mortality in patients with anemia and acute myocardial infarction. ACE: Angiotensin-converting enzyme; CCBs: Calcium channel blockers; CI: Confidence interval; RR: Risk ratio.

Possible advantages of random effects models over fixed effect

i) In theory random effects models accept that true effects differ between studies (fixed does not) and tries to estimate an effect in all possible studies (fixed effects just tries to estimate the effect in the included studies). Many find these theoretical considerations more appealing for random effects.

ii) Intuitively, the more the variability in results between studies, the less certain we are about the pooled estimates of effects. Random effects reflect these uncertainties while fixed effect does not.

Possible advantages of fixed effect models over random effects

i) The gradient in weight between large and small studies is larger in fixed than random models. Since we tend to trust larger studies more than smaller studies, the greater weight that random gives to smaller studies might be perceived as a limitation.

ii) In particular situations when there are large differences between large and small studies random effects sometimes yields results that are not credible.

iii) Since, when there is appreciable heterogeneity, random effects confidence intervals will be wider there is a greater possibility of rating down for imprecision. Thus, in GRADE, when heterogeneity is large, random effects risks double-counting inconsistency in study results: one rates down for inconsistency and then again for precision given the wide confidence intervals driven by the heterogeneity of results.

Reference

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Appendix 3. Calculating absolute effect based on baseline risk and relative effect

Here we show three examples of how one can calculate a risk difference and its associated confidence interval given a relative risk (example 1), an odds ratio (example 2), or a hazard ratio (example 3) and a control group risk. Article 1 in this series provides a glossary explaining the technical terms in this Appendix including absolute risk, relative risk, hazard ratio, risk difference.

Example 1

The meta-analyzed relative risk (RR) of 'disability worsening' in patients with multiple sclerosis from two RCTs (n = 1,479) comparing teriflunomide to placebo proved to be 0.76 (95% CI 0.62 – 0.93).[1]

The risk in the control group (estimated as the median risk in placebo groups) is 21%. When the number of included studies (denoted here as M) is even, the following formula can be used to calculate the median risk:

$$\frac{control\ group\ event\ rate\ in\ \frac{M}{2}th\ study\ +\ control\ group\ event\ rate\ in\left(\frac{M}{2}+1\right)th\ study}{2}$$

Where the studies, M, are sorted by increasing order of event rate.

In case of two control groups (M=2), the median risk is just the average of the first (here, 18.4%) and second risk (here, 23.5%) in the control groups. The median Risk = (18.4% + 23.5%)/2 = 21%.

(1) Risk with intervention (*corresponding risk)* = RR x Risk with control
= 0.76 x 21%
= 16%

(2) Risk difference = Risk with control – Risk with intervention
= 21% – 16%
= 5%

We can use the same procedure to calculate the confidence interval (CI) associated with the risk difference, substituting the extremes of the CI (here, 0.62 and 0.93) for the point estimate (here, 0.76).

(3) For the upper limit of the RR CI (0.93):
- Risk with intervention = 0.93 x 21% = 19.5%
- Risk difference = Risk with control – Risk with intervention = 21% – 19.5% = 1.5%

(4) For the lower limit of the RR CI (0.62):
- Risk with intervention = 0.62 x 21% = 13.0%
- Risk difference = Risk with control – Risk with intervention = 21% – 13.0% = 8.0%

(5) Risk difference is: 5% (95% CI: 1.5% to 8.0%).

Example 2

The odds ratio (OR) of 'incident hypertension' in pregnant women with overweight or obesity from four RCTs (n = 1,324) comparing exercise therapy + usual care to usual care is 0.52 (95% CI 0.28 – 0.96).[2]

The risk in the control group (estimated as the median risk in usual care groups) is 14%. When the number of included studies is even, the formula to calculate the median risk mentioned in example 1 can be applied. In case of four control groups (M=4), the median risk is just the average of the second (here, 9%) and third risk (here, 19.1%) in the control groups. The median risk = (9% + 19.1%)/2 = 14%.

To convert an odds ratio to a RR the next formula[3] can be used:

(1) $RR = \frac{OR}{(1-risk\ in\ control\ group) + (risk\ in\ control\ group \times OR)}$

$= \frac{0.52}{(1-0.14)+(0.14 \times 0.52)}$

= 0.56

We can use the same procedure to calculate to confidence interval (CI) associated with the odds ratio, substituting the extremes of the CI (here, 0.28 and 0.96) for the point estimate (here, 0.52).

(2) For the upper limit of the OR CI (0.96):

- RR = $\frac{0.96}{(1-0.14)+(0.14 \times 0.96)}$ = 0.97

(3) For the lower limit of the OR CI (0.28):

- RR = $\frac{0.28}{(1-0.14)+(0.14 \times 0.28)}$ = 0.31

Now we have the RRs of the point estimate (0.58) and its associated confidence interval (0.31 – 0.97) for a risk in the control group (estimated as the median risk in placebo groups) of 14%.

(4) Risk with intervention (*corresponding risk)* = RR x Risk with control

= 0.56 x 14%

= 7.8%

(5) Risk difference = Risk with control – Risk with intervention

= 14% – 7.8%

= 6.2%

We can use the same procedure to calculate the confidence interval (CI) associated with the risk difference, substituting the extremes of the CI (here, 0.31 and 0.97) for the point estimate (here, 0.56).

(6) For the upper limit of the RR CI (0.97):

- Risk with intervention = 0.97 x 14% = 13.6%
- Risk difference = Risk with control – Risk with intervention = 14% – 13.6% = 0.4%

(7) For the lower limit of the RR CI (0.31):

- Risk with intervention = 0.31 x 14% = 4.3%
- Risk difference = Risk with control – Risk with intervention = 14% – 4.3% = 9.7%

Risk difference is: 6.2% (95% CI: 0.4% to 9.7%).

Example 3

The hazard ratio (HR) of survival in patients with advanced hepatocellular carcinoma (n = 602) comparing sorafenib to placebo is 0.69 (95% CI 0.55 – 0.87).[4] Patients were followed up to about 17 months. We present two calculations. The first calculation relates to *event-free survival*. The second calculation relates to an *event*, i.e. death in this example.

The proportion of event-free patients up to about *eight* months in the control group is 0.50 (50%).
To convert a hazard ratio (HR) to a proportion of event-free patients in the intervention group the next formula[5] can be used:

(1) Proportion with intervention = $e^{\ln\,(control\ group\ event-free\ proportion)\ \times HR}$
= $e^{\ln(0.50)\ \times 0.69}$
= $e^{-0.6931\ \times 0.69}$
= $e^{\ln(-0.4783)}$
= 0.6199 (i.e. 62.0% when rounded)

(2) Event-free survival difference = proportion event-free patients with intervention – proportion event-free patients with control
= 62% – 50% = 12.0% up to about eight months

We can use the same procedure to calculate the confidence interval (CI) associated with the hazard ratio, substituting the extremes of the CI (here, 0.55 and 0.87) for the point estimate (here, 0.69).

(3) For the upper limit of the CI (0.87):
- Proportion with intervention = $e^{\ln(0.50)\ \times 0.87}$ = 0.5471 (i.e. 54.7% rounded)
- Event-free survival difference = proportion event-free patients with intervention – proportion event-free patients with control
  = 54.7% – 50% = 4.7%

(4) For the lower limit of the CI (0.55):
- Proportion with intervention = $e^{\ln(0.50)\ \times 0.55}$ = 0.6830 (i.e. 68.3% rounded)
- Event-free survival difference = proportion event-free patients with intervention – proportion event-free patients with control
  = 68.3% – 50% = 18.3%

(5) Event-free survival difference up to about eight months is: 12.0% (95% CI: 4.7% to 18.3%).

The next section addresses calculating the occurrence of an event, i.e. death in this scenario.

As mentioned earlier, the second calculation relates to the occurrence of an event, i.e. death in this example; we use the same data as used for event-free survival. The proportion of patients with event in the control group [i.e. control group risk] up to about *ten* months is 0.625 (62.5%).

To convert a hazard ratio (HR) to an absolute risk in the intervention group the next formula [5] can be used:

(1) Risk with intervention = $1 - e^{\ln(1-control\ group\ proportion\ events) \times HR}$
= $1 - e^{\ln(1-0.625) \times 0.69}$
= $1 - 0.5083$ = 0.4917 (i.e. 49.2% when rounded)

(2) Risk difference = Risk with control – Risk with intervention
= 62.5% – 49.2%
= 13.3% up to about ten months

We can use the same procedure to calculate the confidence interval (CI) associated with the hazard ratio, substituting the extremes of the CI (here, 0.55 and 0.87) for the point estimate (here, 0.69).

(3) For the upper limit of the CI (0.87):
- Risk with intervention = $1 - e^{\ln(1-0.625) \times 0.87}$ = 0.5740 (i.e. 57.4% rounded)
- Risk difference = Risk with control – Risk with intervention = 62.5% – 57.4% = 5.1%

(4) For the lower limit of the CI (0.55):
- Risk with intervention = $1 - e^{\ln(1-0.625) \times 0.55}$ = 0.4169 (i.e. 41.7% rounded)
- Risk difference = Risk with control – Risk with intervention = 62.5% – 41.7% = 20.8%

(5) Risk difference up to about ten months is: 13.3% (95% CI: 5.1% to 20.8%).

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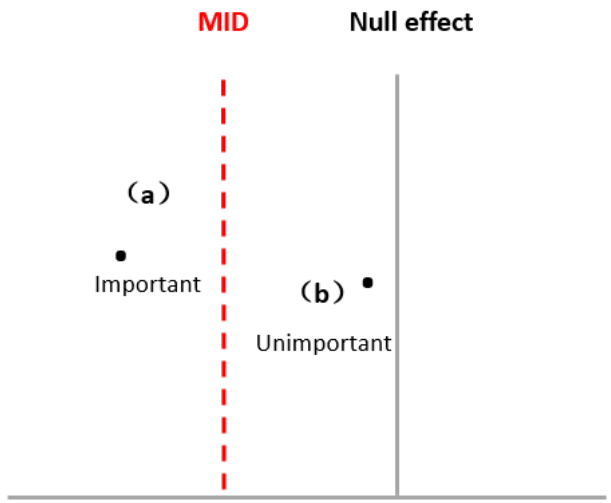
Appendix 4. Challenges and possible solution when one targets the null and the point estimate turns out to be very close to the null

1. The challenge of a point estimate near the null

Core GRADE users who choose the null typically do so at least in part to avoid the value judgements involved in specifying an MID. When the point estimate is not close to the null, they will rate their certainty in a true underlying benefit or harm. When the point estimate is close to the null (b in Figure 1), however, it becomes counterintuitive to rate certainty in a true benefit or harm. The reason is that the most likely truth is little or no effect, and it is therefore most appropriate to rate certainty in that inference.[1]

Figure 1. Target of the certainty rating with respect to the null. If the point estimate falls close to the null as in situation b Core GRADE users should consider revising the target of the certainty rating from a non-zero effect to little to no effect. If Core GRADE users are confident that the point estimate is not close to the null (i.e., larger than the MID, as in situation a), they will continue to rate certainty in a non-zero effect.

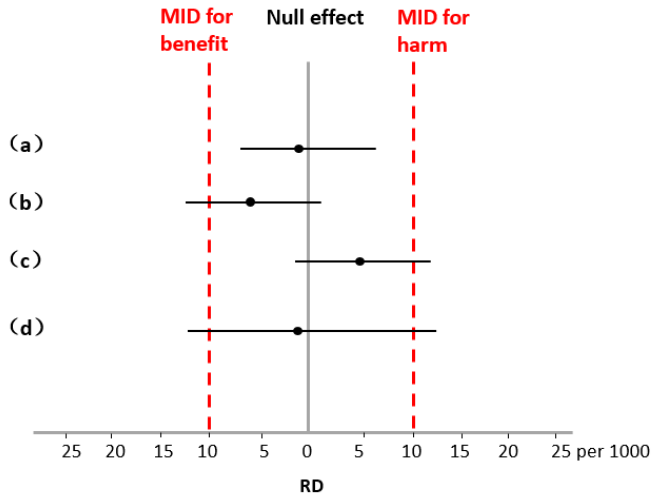


Core GRADE users then face the challenge of deciding what they mean by "little or no effect". The answer is "an unimportant effect" or "an effect less than the MID" (b in Figure 1). They are then in the position they were hoping to avoid: considering peoples' values and making the judgement of what their target audience would consider an MID.

1.1 Deciding whether to rate down for imprecision

Having decided they are now in the position of rating their certainty in little or no effect, it appears on the surface that they must choose a specific single value of the MID. They will then rate down certainty for imprecision if the confidence interval crosses the threshold of the MID for benefit or the MID for harm (b or c in Figure 2) or both (d in Figure 2). If the CI crosses neither threshold they will not rate down for imprecision (a in Figure 2).

Figure 2. Rating certainty in little or no effect. If the CI does not cross the MID (situation a), Core GRADE users will not rate down for imprecision; if crosses, they will rate down (situation b, c, d).



It turns out, however, that Core GRADE users choosing the null as their threshold may avoid having to choose a specific MID. In doing so, they avoid the most challenging of the value judgements they were hoping to forego.

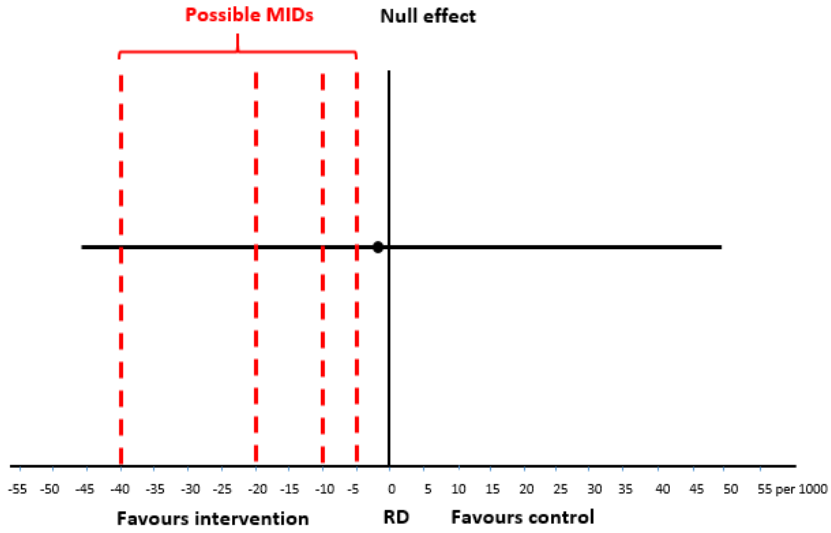
2. Ameliorating the challenge of choosing an MID

Judgments regarding the MID will be less onerous for Core GRADE users if they are required only to consider a range in which the MID might lie rather than a specific MID. The following discussion demonstrates how this is possible.

Core GRADE users who have chosen the null as a threshold and face a point estimate near the null may avoid specifying an exact value for the MID by considering the boundaries of the CI. If those boundaries are clearly within the range that the MID lies (i.e. the ends of the confidence interval represent a lower value than the MID), they needn't rate down for imprecision. If the boundary of the CI is clearly greater than the range in which the MID lies (i.e. one or both ends of the confidence represent a higher value than the MID) they will rate down for imprecision.

For instance, consider a hypothetical example in which the lower boundary of the CI is a reduction in mortality of 50 per 1000 (Figure 3). Without specifying that the threshold of importance is 5, 10, or 20 or 40 in 1,000, all would agree this magnitude represents an important benefit and thus crosses the threshold - wherever exactly that threshold lies. Therefore, without specifying a single specific value of the confidence interval, one would certainly rate down for imprecision.

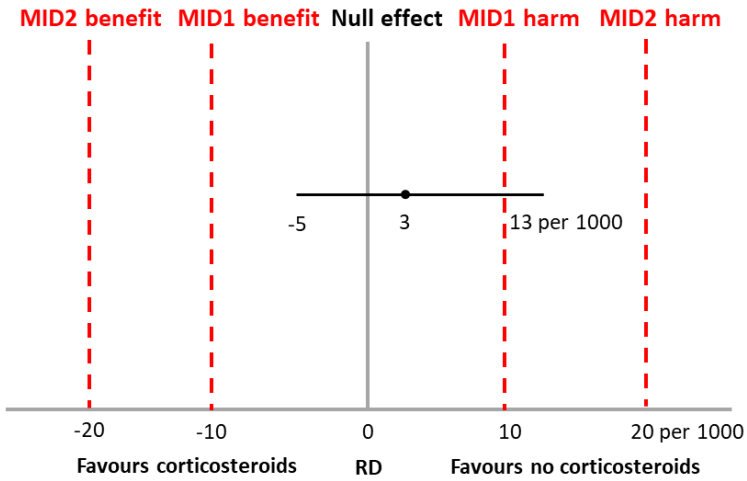
Figure 3. A hypothetical example of imprecision judgment without explicitly specifying an MID threshold in a meta-analysis on mortality. A judgment of whether the boundary of the confidence interval is greater or less than the MID is required. The lower boundary of the confidence interval (a reduction in mortality of 50 per 1000) represents an important benefit whether the MID is a reduction of 5, 10, 20 or 40 events per 1,000 (the broken lines) and thus crosses the threshold warranting rating down for imprecision.



3. Ameliorating the challenge of choosing an MID, an example from a systematic review

Consider a systematic review evaluating the effect of corticosteroid treatment in patients with sepsis. The systematic review, conducted to inform a *BMJ* Rapid Recommendation, included 17 RCTs enrolling over 4,000 participants.[2] It reported a point estimate of 3 additional gastrointestinal bleeds per 1000 patients in those randomized to corticosteroids, with a 95% CI ranging from 5 fewer to 13 more bleeds per 1000 patients (Figure 4).

Figure 4. An example of rating certainty in little to no effect without prespecified MIDs in a systematic review of corticosteroids in sepsis with relation to gastrointestinal bleeding. If the implicit threshold is close to 10 in 1,000 (MID1) or less, Core GRADE users would rate down for imprecision. If it is in the vicinity of 20 in 1,000 (MID2) they would not.



Let us say that the GRADE team had planned to choose the null as the threshold and thus to rate their certainty in a non-zero effect. They now, however, face a point estimate of an increase of 3 bleeds in 1,000, very close to the null and representing an effect that they consider unimportant. Thus, they choose to rate their certainty in little to no effect.

When doing so, they must consider possible estimates of the MID for benefit and harm. Let us say their threshold is close to 10 in 1,000 (MID1 in Figure 4). Since the boundary of the CI favoring no corticosteroid is appreciably greater than 10 in 1,000, they lose confidence in their inference of an unimportant increase in bleeding and would rate down for imprecision.

If, however, they believe the threshold of importance is in the vicinity of 20 in 1,000, the boundaries of the CI are appreciably lower than their thresholds and they would not rate down for imprecision. They would conclude that, at least as far as imprecision is concerned, the data provides compelling evidence that corticosteroids result in little to no increase in bleeding.

This example demonstrates how Core GRADE users choosing the null as the threshold may avoid, when the point estimate is near the null, choosing a single specific value of the MID. In this case, the only judgment review authors need to make is whether the MID is less than the upper limit (harm) of the CI, that is 13 per 1,000 (i.e. 1 to 12 per 1000) or more than 13 per 1,000 (14 or more per 1000). Core GRADE users would rate down for imprecision if they consider the MID less than the upper limit, otherwise not. In describing their decision to rate down or not, authors need to be explicit only in their judgment of whether boundaries of the confidence interval represent important or unimportant effects.

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Appendix 5. Assessing imprecision in the context of a large effect from a meta-analysis of trials with small sample sizes using binary and continuous outcomes

This appendix provides a worked example of, in the context of an implausibly large effect with limited sample sizes, the approach to calculating the optimal information size (OIS) suggested in the main text, as well as alternative approaches to addressing the OIS beyond the actual calculation of optimal information size presented in the main text.

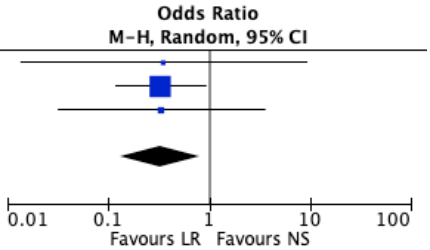
OIS is akin to the concept of the required sample size in individual clinical trials. If the total number of patients included in a systematic review is less than the number of patients generated by a conventional sample size calculation for a single adequately powered trial, Core GRADE users consider rating down the certainty of evidence for imprecision.[1]

Example 1: Consider a systematic review of randomized controlled trials that compared Lactated Ringer's (LR) versus Normal Saline (NS) fluid resuscitation in patients with acute pancreatitis.[2] The meta-analysis of three randomized controlled trials concluded that compared to those who received NS, patients resuscitated with LR were less likely to require ICU admission (OR 0.33; 95 % CI 0.13-0.81). (Appendix 5 Figure 1)

Figure 1. Meta-analysis of randomized controlled trials on Lactated Ringer's (LR) versus Normal Saline (NS) fluid resuscitation for ICU admission in patients with acute pancreatitis

Study or Subgroup	LR Events	LR Total	NS Events	NS Total	Weight	Odds Ratio M-H, Random, 95% CI
de-Madaria 2018	0	19	1	21	7.7%	0.35 [0.01, 9.13]
Lee 2020	6	61	15	60	77.6%	0.33 [0.12, 0.91]
Wu 2011	1	19	3	21	14.7%	0.33 [0.03, 3.51]
Total (95% CI)		**99**		**102**	**100.0%**	**0.33 [0.13, 0.81]**
Total events	7		19			

Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 0.00$, df = 2 (P = 1.00); $I^2 = 0\%$
Test for overall effect: Z = 2.41 (P = 0.02)



Let us assess the imprecision of the pooled effect with the null threshold. While the pooled effect of this meta-analysis appears sufficiently precise, the large effect (67% odds reduction) in the context of only 26 events overall (7/99 in LR versus 19/102 in NS) raises questions about the plausibility of the pooled effect and thus the possible desirability of rating down for imprecision. In other words, the large effect (in relative terms for binary outcomes) in the context of limited sample size and number of events triggers the need to address the OIS. We review here possible ways that Core GRADE users might go about addressing the issue.[1 3 4]

Option 1: Using hypothesis testing approach to calculate OIS

OIS can be calculated using the following steps:

Step 1. Determine the expected effect size: Estimate the plausible magnitude of the effect that the intervention is anticipated to have. For OIS we typically use relative risk reductions (RRR) of 20% or 25% as the expected effect size.

Step 2. Set the desired power and significance level: Typically, we use a power of 80% and a significance level of 5%.

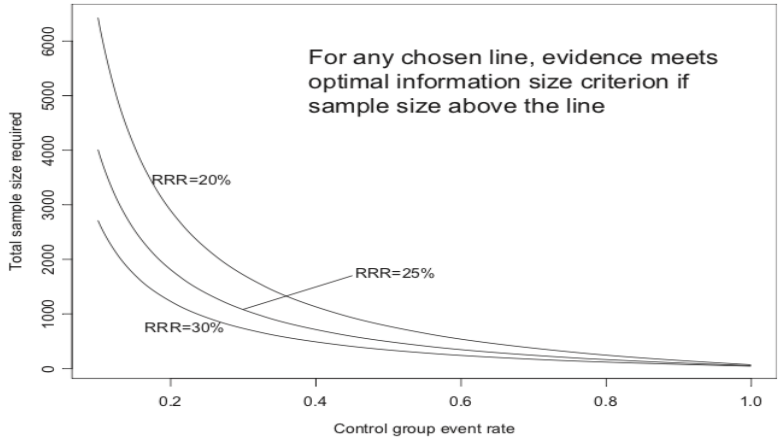
Step 3. Use standard sample size formulas: Employ standard formulas for calculating sample size based on the type of outcome measure (e.g., binary, continuous). Many online calculators for sample size calculation are available. One such online calculator is available at https://www.openepi.com/SampleSize/SSCohort.htm.

For the systematic review on RL versus NS, let us calculate the OIS for the outcome ICU admission using a RRR of 25% as the expected effect size, with proportions with the outcome in intervention and control arm as 9% and 12% respectively. The OIS using these assumptions would be a total of 3406 (1703 in each arm), which was very far from being met in the meta-analysis (total number of patients across all trials 201). This would prompt authors to rate down for imprecision in the GRADE assessments.

Option 2: Using figures to determine OIS and minimum number of events

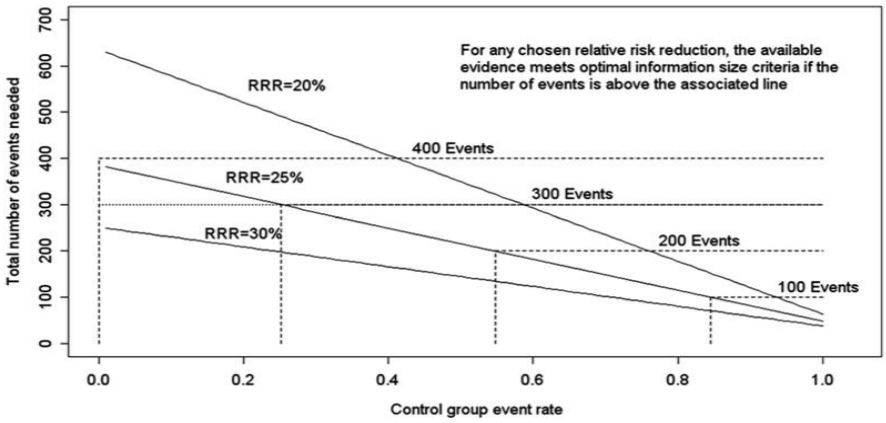
As an alternative to calculating the OIS, review and guideline authors can refer to a figure to determine the OIS.[3] Appendix 4 Figure 2 shows the required sample size (assuming an alpha of 0.05 and a beta of 0.2) for relative risk reductions (RRR) of 20%, 25%, and 30% across various control event rates. In the pancreatitis example, if the estimated control event rate is 0.12 and an RRR of 25% is specified, the OIS is approximately 3500 patients. At the same control event rate, with a RRR of 20%, the OIS is approximately 6400, and with a RRR of 30%, the OIS is approximately 2500. Given that the total number of patients across all trials in the pancreatitis meta-analysis was only 201, which is way below the OIS indicated by these figures, authors would be urged to rate down for imprecision in the GRADE assessments.

Figure 2. Optimal information size given a of 0.05 and b of 0.2 for varying control event rates and relative risks.[3]



However, power is more closely related to the number of events than to sample size. Appendix 4 Figure 3 illustrates the same relationships using the total number of events across all studies in both the treatment and control groups instead of the total number of patients. Keeping the same assumptions as in the prior paragraph (control event rate 0.12 and RRR 25%), one requires approximately 360 events to meet OIS criteria. Similarly, keeping the control events rate at 0.12, a RRR of 20% and 30% would require approximately 630 and 250 events respectively to meet OIS criteria. The total number of events across all trials in the pancreatitis example is only 26, far below the OIS. This would prompt authors to rate down for imprecision.

Figure 3. Optimal information size presented as number of events given a of 0.05 and b of 0.20 for varying control event rates and RRR of 20%, 25% and 30%. Abbreviation: RRR, relative risk reduction.[3]



Option 3: Assessing if the ratio of the upper to the lower boundary of the CI is more than 2.5 for odds ratio (OR) or 3 for Risk Ratio (RR)

The ratio of upper and lower boundary of the CI may indicate if the sample size in the meta-analysis is less than the OIS if it is greater than 2.5 for OR and 3 for RR.[4] This may in fact preclude the need to carry out a calculation of OIS for the meta-analysis. For the example above, a relative effect of 67% reduction in odds of patients requiring ICU admission represents a large, probably implausible effect with a small sample size of 201 (number of events=26), and systematic reviewers and guideline developers would therefore want to determine whether the OIS is met.

When, as in this situation, the sample size or number of events is far lower than the OIS, Core GRADE users should consider rating down twice for imprecisions. The criterion we suggest for this decision is to look at the ratio of the upper CI boundary to the lower CI boundary: if that ratio is larger than 2.5, we suggest to rate down twice. [1] In this case the number of events is almost an order of magnitude less than the OIS, and the of the upper boundary of the CI (0.81) to the lower CI boundary (0.13) for the OR is 6.2 which is larger than 2.5. This suggests the appropriateness of rating down two levels for imprecision.

Example 2: Consider a systematic review of randomized controlled trials for stem cell therapy for chronic liver disease.[5] Appendix 4 Figure 4 depicts the results of the meta-analysis for the outcome MELD score MELD (Model for End-Stage Liver Disease) at 8 weeks, arising from four trials with a total of 248 participants studying stem cell therapies versus standard of care. The score is a numerical scale used to assess the severity of chronic liver disease and prioritize patients for liver transplantation, with a higher score indicating a more severe liver disease and a greater urgency for liver transplantation. For dichotomous outcomes, it is recommended to rate down two levels if the upper-to-lower CI ratio exceeds 2.5 for odds ratios or 3 for risk ratios. For continuous outcomes, rate down two levels if the total sample size is less than 30–50% of the optimal information size. [1]

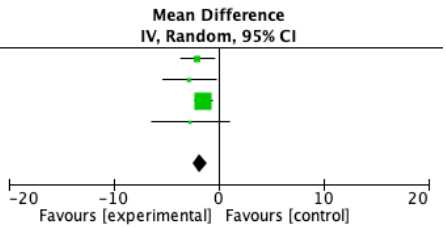
Let us assess the imprecision of the pooled effect with the threshold being an MID of 4. The confidence interval (-2.45 to -1.05) seems sufficiently precise with respect to this threshold. If we would rate the certainty that stem cell therapy had an important reduction in MELD score (i.e., the target of certainty rating), we would not rate down for imprecision.

One can make a calculation of OIS for this outcome by specifying the MID and the standard deviation from one of the existing studies. Alternatively, a rule of thumb would be to not rate down for imprecision if there are a minimum of 400 patients per group (i.e., a total of 800 patients). Since the given meta-analysis has a total of only 248 patients across all trials, this would prompt us to rate down for imprecision using these criteria.

Figure 4. Meta-analysis of randomized controlled trials on stem cell therapy for reduction in MELD score in chronic liver disease

	Experimental			Control				Mean Difference
Study or Subgroup	Mean	SD	Total	Mean	SD	Total	Weight	IV, Random, 95% CI
Amer ME 2011	11.37	2.6	20	13.43	2.59	20	18.8%	-2.06 [-3.67, -0.45]
Li YY 2015	18.54	5.27	31	21.38	4.59	27	7.6%	-2.84 [-5.38, -0.30]
Xu WX 2019	14.97	2.6	60	16.47	1.43	30	70.1%	-1.50 [-2.33, -0.67]
Zekri AR 2015	21.43	7.3	30	24.17	7.35	30	3.5%	-2.74 [-6.45, 0.97]
Total (95% CI)			**141**			**107**	**100.0%**	**-1.75 [-2.45, -1.05]**

Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 1.47$, df = 3 (P = 0.69); $I^2 = 0\%$
Test for overall effect: Z = 4.92 (P < 0.00001)



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